

1. Administrative & Study Identification

Full Study Title: Phase 3 Study of Pembrolizumab in Combination With Carboplatin/Taxane (Paclitaxel or Nab-paclitaxel) Followed by Pembrolizumab With or Without Maintenance Sacituzumab Tirumotecan (Sac-TMT; MK-2870) in the First-line Treatment of Metastatic Squamous Non-small Cell Lung Cancer [MK-2870-023]
Short Title/Acronym: MK-2870-023 **Protocol Number:** MK-2870-023 **NCT Number:** NCT06422143 **Posting ID:** 1029816075870010455 **Trial Status:** RECRUITING **Sponsor Name:** Merck Sharp & Dohme LLC (Company: Merck) **Investigational Product:** Pembrolizumab, Sacituzumab Tirumotecan (Sac-TMT; MK-2870), Carboplatin (Trade Name: Paraplatin), Paclitaxel, Nab-paclitaxel **Phase of Development:** Phase III **Medical Monitor:** Merck Clinical Operations Lead [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether pembrolizumab with maintenance sacituzumab tirumotecan (Sac-TMT) is superior to pembrolizumab without sacituzumab tirumotecan maintenance with respect to overall survival (OS). **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), the number of participants with ≥ 1 Adverse Event (AE), and the number of participants discontinuing from study therapy due to AE(s).

2.2 Background & Rationale To address the gap in long-term survival outcomes for patients with metastatic squamous non-small cell lung cancer (NSCLC). This study evaluates the efficacy of adding a novel antibody-drug conjugate (sacituzumab tirumotecan) to standard-of-care pembrolizumab maintenance therapy following a first-line induction phase of chemo-immunotherapy, aiming to improve overall survival in this patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab maintenance alone) **Blinding:** Open-label / Randomized **Randomization:** Randomized (Following induction phase of four cycles, participants are randomly assigned to pembrolizumab maintenance vs. pembrolizumab + sac-TMT maintenance).

3.2 Planned Enrollment & Duration Trial Status: RECRUITING **Target N\$:** 851 participants **Number of Sites:** Global study (approx. 205

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of squamous non-small cell lung cancer (NSCLC) [Stage IV: M1a, M1b, M1c, American Joint Committee on Cancer Staging Manual, version 8]. 2. Measurable disease per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 as assessed by the local site investigator/radiology. 3. Has a life expectancy ≥ 3 months and an Eastern Cooperative Oncology Group Performance Status (ECOG PS) score of 0 or 1 assessed within 7 days prior to allocation. 4. Archival tumor tissue sample or newly obtained core, incisional, or excisional biopsy of a tumor lesion not previously irradiated has been provided. 5. Well-controlled HIV on antiretroviral therapy (ART) allowed; HBsAg-positive eligible if on HBV antiviral therapy for ≥ 4 weeks with undetectable viral load; history of HCV eligible if viral load is undetectable at screening. 6. AEs due to previous anticancer therapies recovered to Grade ≤ 1 or baseline (participants with endocrine-related AEs adequately treated with hormone replacement and those with Grade ≤ 2 neuropathy are eligible). Has adequate organ function. 7. **For Maintenance only (prior to randomization):** Without disease progression of NSCLC determined by BICR using RECIST 1.1 after study-specified Induction (evaluative scan at Week 12); has ECOG PS 0 or 1 at Prerandomization Visit; all AEs (except alopecia, Grade 2 fatigue, and Grade ≤ 2 endocrine AEs requiring tx) have recovered.

4.2 Exclusion Criteria 1. Diagnosis of small cell lung cancer or, for mixed tumors, presence of small cell elements. 2. Received prior systemic chemotherapy or other targeted/biological antineoplastic therapy for their metastatic NSCLC (prior neo/adjuvant or chemoradiation for nonmetastatic allowed if completed ≥ 12 months before metastatic diagnosis). 3. Prior therapy with anti-PD-1, anti-PD-L1, anti-PD-L2, or agents directed to other stimulatory/co-inhibitory T-cell receptors (e.g., CTLA-4, OX-40, CD137) for metastatic disease. Prior TROP2-targeted ADC or topoisomerase I inhibitor-containing ADC. 4. Uncontrolled, significant cardiovascular/cerebrovascular disease (e.g., NYHA Class III/IV, unstable angina, myocardial infarction, uncontrolled symptomatic arrhythmia, QTcF interval >480 ms) within 6 months. Known CNS metastases/carcinomatous meningitis. 5. History of documented severe dry eye syndrome, severe Meibomian gland disease, and/or blepharitis, or severe corneal disease. 6. Active inflammatory bowel disease requiring immunosuppressive medication or previous history (e.g., Crohn's, ulcerative colitis, chronic diarrhea). 7. HIV-infected with newly diagnosed/history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. Known additional malignancy progressing or

requiring active tx within past 3 years. 8. Lung radiation ≥ 30 Gray within 6 months; prior radiotherapy within 2 weeks; or radiation-related toxicities requiring corticosteroids. 9. Severe hypersensitivity ($\geq$ Grade 3) to study intervention/excipients or biologic therapy. Active autoimmune disease requiring systemic treatment past 2 years. History of or current (noninfectious) pneumonitis/ILD requiring steroids. Active infection requiring systemic therapy. History of allogeneic tissue/solid organ transplant. 10. Received live/live-attenuated vaccine within 30 days before first dose; unrecovered from major surgery/ongoing complications; currently receiving strong inducer/inhibitor of CYP3A4 that cannot be discontinued (2-week washout required).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Induction Phase* (4 cycles): Pembrolizumab (q3w) + Carboplatin [Paraplatin] (q3w) + Paclitaxel (q3w) or Nab-paclitaxel (weekly). * *Maintenance Phase:* Pembrolizumab (q3w) + Sac-TMT vs. Pembrolizumab (q3w) alone. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or study completion (up to ~50 months).

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care therapies and physiological hormone replacement therapy (e.g., thyroxine, insulin, or physiologic corticosteroids). **Prohibited:** Systemic immunosuppressive medications for active inflammatory bowel disease, concurrent investigational agents, strong inducers/inhibitors of CYP3A4 without proper washout, live-attenuated vaccines within 30 days of the first dose, and prior topoisomerase I inhibitor-containing ADCs or TROP2-targeted ADCs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening: Informed Consent, Medical History, Tumor Tissue sample provision, Baseline Imaging (RECIST 1.1)
2-5	Cycles 1-4	Induction: Pembrolizumab + Carboplatin + Paclitaxel/Nab-paclitaxel administration
6	Week 12	Randomization: RECIST 1.1 evaluation
7-50	Q3W	Maintenance: Pembrolizumab or Sac-TMT administration, Safety Labs, Efficacy Assessment
51	Up to ~50 Mos	End of Study: Final Survival Follow-up, Safety Monitoring, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS) - defined as the time from randomization to death due to any cause.

Measurement Method: Clinical mortality tracking and survival analysis.

7.2 Secondary Endpoints [Endpoint 1]: Progression-Free Survival (PFS) **[Endpoint 2]:** Number of Participants With ≥ 1 Adverse Event (AE) **[Endpoint 3]:** Number of Participants Discontinuing from Study Therapy Due to AE(s)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during study visits, tracking toxicities including severe dry eye syndrome, pneumonitis/ILD, and infusion-related reactions. Secondary endpoints explicitly track AE occurrences and resulting study discontinuations. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External committee per Merck Sharp & Dohme LLC standard safety protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary OS/PFS outcomes; Safety Set for adverse events. **Sample Size Justification:** Powered to detect a statistically significant superiority in Overall Survival (OS) for the pembrolizumab + sac-TMT maintenance arm compared to pembrolizumab alone among 851 enrolled participants. **Handling of Missing Data:** Standard survival analysis methodologies (e.g., right-censoring for OS and PFS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standardized onsite and remote clinical monitoring per sponsor guidelines. **Audit Trail:** eCRF locking and blinded independent central review (BICR) for RECIST 1.1 progression evaluations.

11. Study Identifiers & Governance

Primary ID: MK-2870-023 **Posting ID:** 1029816075870010455 **Registry Number:** NCT06422143 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (Company: Merck) **Trial Status:** RECRUITING **Study Title:** Pembrolizumab With or Without Maintenance Sacituzumab Tirumotecan (Sac-TMT; MK-2870) in Metastatic Squamous Non-small Cell Lung Cancer (NSCLC) [MK-2870-023] **Official Regulatory Title:** Phase 3 Study of Pembrolizumab in Combination With Carboplatin/Taxane

(Paclitaxel or Nab-paclitaxel) Followed by Pembrolizumab With or Without Maintenance MK-2870 in the First-line Treatment of Metastatic Squamous Non-small Cell Lung Cancer

12. Clinical Framework (Oncology Specific)

Primary Condition: Non-small Cell Lung Cancer, NSCLC / Squamous Cell Carcinoma of Lung **Diagnosis Threshold:** Stage IV (M1a, M1b, M1c) per AJCC version 8 **Medical Methodology:** Induction Chemo-immunotherapy followed by Maintenance therapy **Optimization Target:** Superior Overall Survival (OS) and Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: Standard first-line oncology treatment pathway adapted with novel ADC maintenance **Education Protocol (HCP):** Site initiation visits and RECIST 1.1 centralized training **Education Protocol (Patient):** Informed consent process, infusion management education, and AE self-reporting guidance **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for disease progression

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 50 months (Primary Completion Target: Jan 2029) **Onsite Visit Cadence:** Every 3 weeks (Q3W) matching infusion cycles **Remote Monitoring Frequency:** Standard clinical follow-up between infusion cycles **Checklist Review Interval:** Regular tumor imaging/assessments (e.g., Week 12 and subsequently as per protocol)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				